

7.4.29

EE25BTECH11022 - sankeerthan

Question: If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord to the circle with the center $(2, 1)$, then the radius of the circle is

- 1) $\sqrt{3}$
- 2) $\sqrt{2}$
- 3) 3
- 4) 2

Solution: Given the circle equation:

$$x^2 + y^2 - 2x - 6y + 6 = 0 \quad (1)$$

$$\mathbf{x}^T \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0, \text{ where } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} g \\ f \end{pmatrix} \quad (2)$$

From the given equation:

$$2\mathbf{u}^T \mathbf{x} = -2x - 6y \implies \mathbf{u} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}, \quad f = 6 \quad (3)$$

Center of the circle is:

$$\mathbf{c}_1 = -\mathbf{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \quad (4)$$

Radius of the circle is:

$$r_1 = \sqrt{\mathbf{u}^T \mathbf{u} - f} = \sqrt{(-1)^2 + (-3)^2 - 6} = \sqrt{4} = 2 \quad (5)$$

Given the second circle has center:

$$\mathbf{c}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad (6)$$

Assuming the diameter chord line of the first circle with equation $y = 3$,

$$\mathbf{n} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad c = 3 \quad (7)$$

Distance from second circle center $\mathbf{c}_2$ to this chord:

$$\frac{|\mathbf{n}^T \mathbf{c}_2 - c|}{\|\mathbf{n}\|} = \frac{|1 - 3|}{1} = 2 \quad (8)$$

Therefore, the radius of the second circle $r=2$.

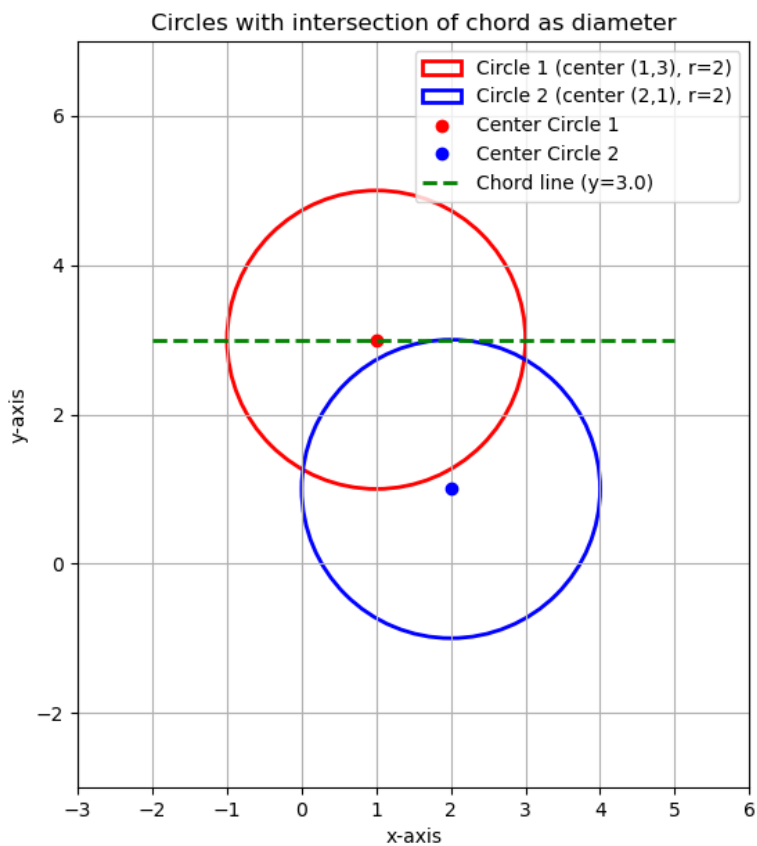


Fig. 4.1